MONASH University

ETW3482

Week Five Tutorial



This week's tutorial will have **two sessions**

1. **Test Your Understanding of Week 5 Content**
2. **Demo session: Upload an Excel table to SAS Viya and explore the tree-based models in Model Studio.**



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First Session: Quizzes



Do you dare to take the challenge?

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Which of the following is the best reason for predictive modelling to be a supervised learning method?

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Which of the following option describes the overfitting models correctly?

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What is the main purpose of data partitioning?

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Decision tree models are universal approximators.

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What does "Selecting useful inputs" mean when building a Decision Tree?

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Which of the following techniques is related to "Optimizing complexity" in Decision Trees?

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In the context of Decision Trees, which option best describes the process of "Predicting new cases"?

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In a decision tree model, what is the most likely effect of increasing the maximum tree depth?

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Which of the following methods adaptively reweights subsequent samples based on the residuals from previous combinations of models?

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Which method trains decision tree models from samples obtained by sampling both rows and columns of the training data?

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Second Session: Hands-on activities



Let’s get started!

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Demo session

- Download the **comms_exdata.xlsx** table from the Real-time section on the Week 5 Moodle page.
- Identify the location of the Excel table and open the Excel table.
- Examine the column names. Correct the column names if they do not agree with the SAS naming convention.

1. **Start with a Letter or Underscore:** Variable names must start with a letter (A-Z) or an underscore (_).
2. **Length Limit:** Variable names can be up to 32 characters long.
3. **Characters Allowed:** Use only letters (A-Z), digits (0-9), and underscores (_). Spaces and special characters like $, %, #, etc., are not allowed.

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Which variable names do not agree with the SAS naming convention? Why?

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Demo session

- Follow the tutor’s instructions with the “Uploading an Excel Data to Model Studio Project”.

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When importing a file into SAS Model Studio, which library should you select in the Target location step before proceeding with the import?

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Demo session

- Refer to the Tutorial Demo Guide 1. Building a Decision tree Model with Default Setting, and 2. Improving a Decision Tree Model by Changing the Tree Structure Parameters, and follow the tutor’s instructions.
- Now, let’s go through the demo and answer the following questions.

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What are the implications of increasing the depth and minimum leaf size of a decision tree?

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Demo session

- Refer to the Tutorial Demo Guide 3. Improving a Decision Tree Model by Changing the Recursive Partitioning Parameters, and follow the tutor's instructions.
- Now, let's go through the demo and answer the following questions.

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What do the parameters for recursive partitioning in decision trees primarily control?

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Demo session

- Refer to the Tutorial Demo Guide 4. Improving a Decision Tree Model by Changing the Pruning Parameters, and following the tutor’s instructions.
- Now, let’s go through the demo and answer the following questions.

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What is the primary purpose of pruning parameters in a decision tree model?

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Demo session

- Refer to the Tutorial Demo Guide, 5. Building a Gradient Boosting Model and 6. Building a Binary Target with a Forest, and follow the tutor's instructions.
- Now, let's go through the demo and answer the following questions.

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In Gradient Boosting (Model Studio), you decide to reduce the learning rate significantly to make the model's steps more conservative. To keep model capacity roughly the same, which adjustment is most appropriate?

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In a Forest model, how can you estimate error during training without creating a separate validation set?

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Q&A



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Audience Q&A

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